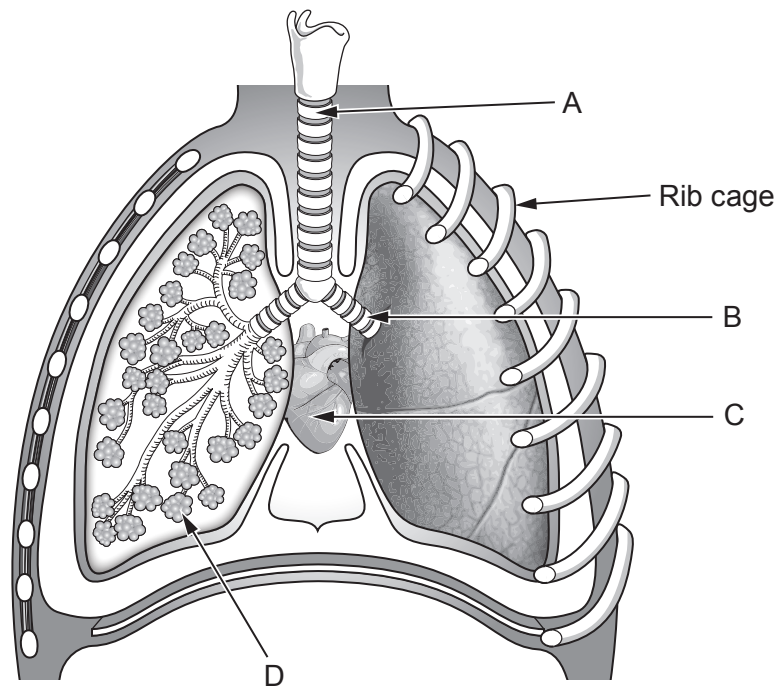


3. (a) **Image 3.1** is a vertical section through the human chest cavity (thorax).

**Image 3.1**



Choose the letter (**A–D**) from **Image 3.1** which shows

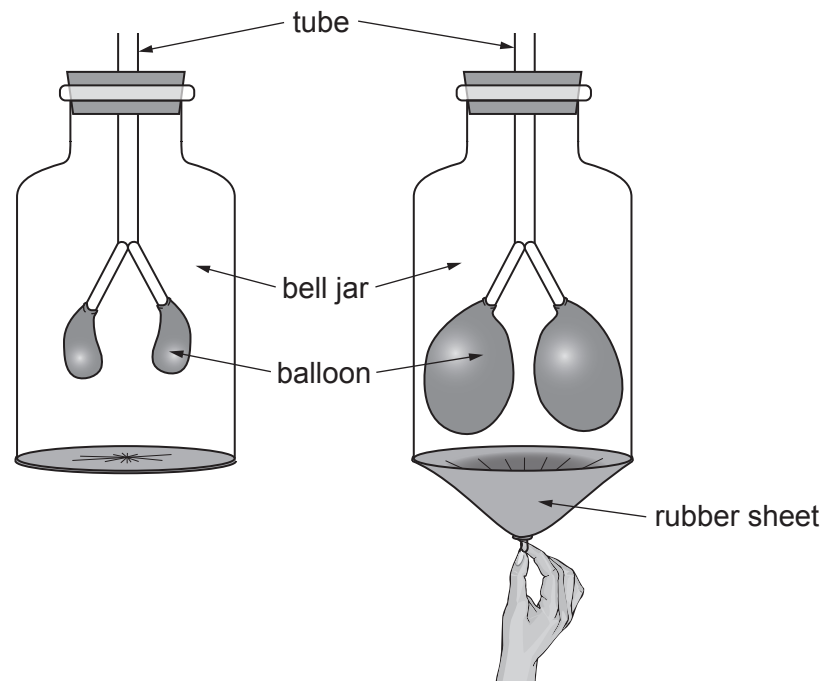
[2]

- (i) a bronchus .....
- (ii) an alveolus .....



- (b) The way in which the respiratory system works during breathing can be demonstrated using the bell-jar model shown in **Image 3.2**.

**Image 3.2**



- (i) State the part of the model shown in **Image 3.2** which represents: [3]
- I. the diaphragm .....
  - II. the lungs .....
  - III. the trachea .....
- (ii) The rubber sheet can be pulled downwards to demonstrate **inspiration** (breathing in).

Complete each of the sentences below by circling the correct statement to describe the process of **inspiration** in the human thorax. [2]

The volume of the thorax      **increases** / **decreases** / **stays the same**.

The pressure in the thorax      **increases** / **decreases** / **stays the same**.



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only

(iii) Describe how the ribcage moves during **inspiration** and why this cannot be demonstrated using the bell jar model. [2]

.....

.....

.....

.....

(c) Human muscle cells usually carry out aerobic respiration. During strenuous exercise, however, anaerobic respiration occurs.

State the harmful chemical substance produced by anaerobic respiration in human muscle cells. [1]

.....

|    |
|----|
|    |
| 10 |

